

科技部補助專題研究計畫成果報告 期末報告

小型手持裝置螢幕之版面留白對閱讀績效和美感喜好的影響

計畫類別：個別型計畫
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中文摘要：文章版面上的留白可以給讀者有種喘息沒壓迫感。平面或綫頁設計師大都鼓吹利用留白元元素於版面設計；然，留白的優點一直停留在設計師們的主觀意見，但現今的研究却少有以科學方法探討留白效應對手機文章閱讀績效的影響。因此，本計畫欲以智慧型手機為實驗工具，探討文字排版的留白效應對閱讀績效和美感喜好的影響。文章排版的留白包含三種：頁留白、行距和段距。本計畫採3(頁留白) $\times$ 2(行距) $\times$ 2(段距)，受測者內隨機因子設計，共有12種水準組合。受測者執行校對測驗，即每位受測者執行12篇國小4年級的閱讀測驗題，一篇文章對應到一種水準組合。每篇閱讀測驗的總字數皆為501字，每篇各有10個錯字。以校對時間和主觀喜好度為效標，即校對時間愈短且美感分數愈高者，即為最佳的留白設計。最後，以ANOVA檢定各因子效應以及其交互作用。實驗結果發現，在閱讀時間方面，頁留白、行距和段距等三個主效應皆顯著，但其交互作用皆不顯著。數據顯示，過大的頁留白面積以及行距，皆不利閱讀績效。然而，增加段距則有利閱讀績效。再者，實驗結果顯示，頁留白對主觀的評量有顯著效應，數據顯示，適當的頁留白(即1/3頁面留白)有利閱讀性，但過大的頁留白面積會提高成閱讀用心度，減少舒適性、美感和整體的喜好度。另外，與客觀績效不同，數據却呈現加大行距皆會提高受測者主觀的舒適感和美感美數。此結果再度說明，主觀喜好和使用性績效不在同一向度，排版設計應同時考量這二項因素。

中文關鍵詞：繁體中文，排版，留白，行距，段距，頁留白，閱讀時間，美感，閱讀性

英文摘要：Chinese characters have substance and occupy space. The empty space is known as negative space, called "White Space" in traditional two-dimensional print design. White space provides relief to readers in reading a tedious text, and reduces loading of the reading task; consequently, improves visual communication and reading comprehension. However, the scientific evidences for the advantages of white space are still barren. Hence the purpose of the study was to explore the effects of white space on readability. The experiment was a 2(line spacing) $\times$ 2(paragraph spacing) $\times$ 3(page spacing) Latin square design with two replicates. Objective performance measure was reading time; Subjective performance measures included subjective ratings of comfort, appealing, preference and effortless. Twenty four undergraduate students from the Medical Colleges in Taiwan were recruited to participate in the experiment. Each subject performed only one experimental run. Results of analysis of variance indicated that the main effects of Page Spacing, Line Spacing and Paragraph Spacing significant on reading time. However, the interactions of those factors were not significant. Besides, the results showed that the effect of Page Spacing

was significant on subjective effort ratings; and both effects of Page Spacing and Line Spacing were significant on subjective comfort ratings; both effects of Page Spacing and Line Spacing were significant on subjective appealing ratings; and Page Spacing was significant on subjective preference ratings. The outcomes demonstrated that oversized white space would degrade readability.

英文關鍵詞：Typography, White Space, Traditional Chinese characters, Line Spacing, Page Spacing, Paragraph Spacing, reading time, aesthetics, readability

1. Introduction

Due to the convenience of handy carrying, users can use smartphones at any time, place or occasions. The penetration rate of smartphone users is still growing up rapidly. According to eMarketer's estimates (2016), 73.4% of Taiwan's population use smartphones. Smartphones are no longer cutting-edge communication gadgets but are now necessities in peoples' lives. Global Views Survey Research Center (2014) reported that 46.4 percent of the Taiwan population had reading experiences on e-Book and that 27.1 percent of them used a smartphone to read the e-Book. The population of reading either digital news or e-books on smartphones would continuously increase due to increasing penetration of smartphones in Taiwan. Most Taiwanese used to reading article written with Traditional Chinese characters. Therefore, it is necessary to explore the effect of typography of Traditional Chinese text on reading performances.

Traditional Chinese characters have substance and occupy space. The space occupied by characters is known as positive space; unoccupied, or empty space is known as negative space (Wong, 1993). The negative space in traditional two-dimensional print design is called "White Space". White space is a frequently used term in graphic and web design, and is applied to graphic or visual communication design. The white space of typographic design includes margin, line spacing and paragraph spacing, etc.

Keyes (1993) proposed a concept of typographic tonality which referred to "the relative blackness or shades of gray of type on the page". Keyes (1993) indicated that typographic tonality described "the way we perceptually group individual character strokes, letters, words, or lines of type to form an overall gray-tone varying from light to dark". That is, tonality refers to the ratio of the area of positive space and that of white space. He believed the low tonality would increase text legibility and comprehension. Furthermore, visually speaking, white space provides relief to readers in reading a tedious text, and reduces loading of the reading task. Golombisky and Hagen (2010) argued that the white space would improve visual communication and reading comprehension. Therefore, the white space would not waste the screen space

However, except line spacing, those suggestions of applying white space for typographic design were from professional experiences of designers, or practitioners; they did not offer any scientific evidences to support their claims. This paper would explore the effect of white space of typography on reading performance on smartphones.

Line spacing

Line spacing, referring to the distance between lines of articles, is a kind of white space, i.e. negative space. Previous studies on reading English words found that the double-line spacing had advantage on computer screens because it decreases lateral masking (Kolers, Duchnicky, & Ferguson, 1981; Morrison and Inhoff, 1981). However, Ganayim, Israel and Ibrahim (2013) found

no significant effect of interline spacing on reading Arabic articles. It might be that the effect of line spacing was different for varied writing systems. It is not appropriate to generalize the findings of those studies to typographic design of Traditional Chinese texts.

Like alphanumeric typography, The large enough line spacing would reduce confusion in location of character positions during reading; line spacing less than 2 times character height did not had advantage on desktop computer screens. For example, Chan and Lee (2005) found that double line spacing on a 15-in displays produced faster reading times and higher preferences than single line spacing because the line spacing of 2-times character height was large enough. In addition, previous studies on reading Chinese characters also found that wider enough line spacing led to better accuracy and less times on reading tasks (Ling and Schaik, 2007), and proofreading tasks (Alan. et al., 2014) in desktop displays. For reading in mobile phones, Wang et al. (2009) also found that participants subjectively preferred large line spacing to small one on readability, fatigue and preference; but they found no significant effect of line spacing on reading time (Wang et al., 2009). However, we did not know if double line spacing had better readability, or faster reading speed than single line spacing because their study only investigated the difference between line spacing sizes which were not larger than 1 characters height.

Besides, the viewing distances of those studies were confined to a fixed distance in their experimental trials. However, users can adjust their viewing distances in the real situations. Therefore, it might be not appropriate to generalize the findings of previous studies to the typographic design of Traditional Chinese texts on smartphone appropriately. For simulating the real situations, the present study did not set a fixed viewing distance and encouraged subjects to adjust their viewing distances in anytime to attain their optimally comfortable viewing situations.

Paragraph spacing

Paragraph spacing referred to the space size between the last line of one paragraph and the first line of the next. Paragraph spacing could chunk continuous text into manageable units according to subject matter. Keyes (1993) argued that the chunking would reduce the typographic tonality and facilitate reading comprehension. Soegaard (2017) also believed the Paragraph spacing may enhance the overall readability and flow of the text. However, they did not offer any scientific evidences to support their claims. Therefore, this article would verify that the appropriate white space between paragraphs could keep each page from looking busy; consequently, readers could reduce their frustration when reading on smartphones.

Page Spacing

In addition, page spacing, a kind of white space, refers to the margin from top of the page to the beginning of the text. For instance, the large white space, shown in Figure 1, is called as page spacing in this current study. Sometimes, an illustration would be put on the page spacing as shown on Figure 1.b. Page spacing is applied to typography of the first page of texts, or of

children book frequently. It might make the screen look concise and provide relief to readers in reading a text. Hence, the present study tried to explore if the page spacing would positively affect reading performances.

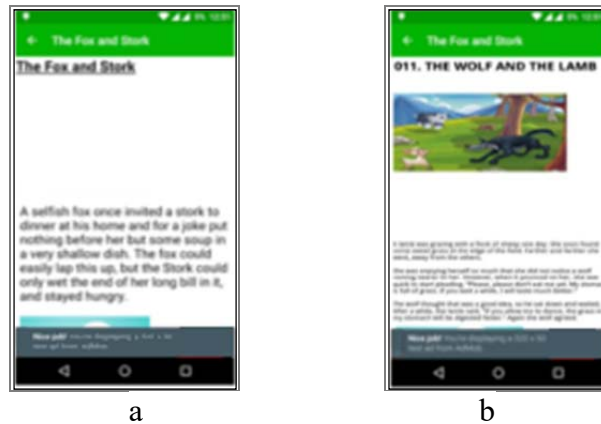


Figure 1. Examples of Page Spacing referring to the margin from top of the page to the beginning of the text.

Scrolling effect

Conversely, page spacing would also reduce the amount of information on a page, which might decelerate reading speeds. Besides, increasing the area of white space would result in that the text would overflow into multiple screens. Users have to scroll the screen in reading all the time, which might negatively affect performance (Morrison & Duncan, 1988; Piolat et al., 1997). Previous studies believed that scrolling would reduce reading comprehension on full-size displays (Sanchez and Wiley, 2009), and eliminate remembering of text-reading on small displays (Sanchez and Goolsbee, 2010). The reason might be that scrolling text would change the locations of all characters in the text, resulting in difficulty for users in locating information by recalling their spatial locations (Morrison & Duncan, 1988). Hence, white space might degrade reading performance because the increase of scrolling behaviors.

2. Methods

Applying white space to typography was advocated by professional designers, or practitioners. However, white space might also degrade reading performance because increasing scrolling behaviors. Therefore, the paper would investigate if the positive effects of white space exist on readability.

The experiment was a 2(line spacing) $\times$ 2(paragraph spacing) $\times$ 3(page spacing) Latin square design with two replicates. The line spacing size had two levels: one font height, and double font height. One font height refers to the height of white space between the successive lines is equal to the height of the font used in the text; double font height refers to the height of white space between the successive lines is twice the height of the font used in the text. Leading, referring to the distance between the baselines of successive lines of text, is used to express the line spacing frequently.

However, the amount of one leading varies depending on the font used; and the true height of the lines of text will be roughly 115% to 120% of the point size. Therefore, the present paper did not use leading to express line spacing; but use the font height to express the height of white space between the successive lines.

The paragraph spacing size also included two levels: one line-spacing height, and double line-spacing height. One line-spacing height refers to the height of white space between the successive paragraphs is equal to the height of the line spacing of the text; for example, the size of Paragraph Spacing would be 2-Character height when the size of Line spacing of the text was 2-Character height. Double line-spacing height refers to the height of white space between the successive paragraphs is twice the height of the line spacing of the text; for example, the size of Paragraph Spacing would be 4-Character height when the size of Line spacing of the text was 2-Character height. When the paragraph spacing is one line-spacing height, it means that the height between line spacing and paragraph spacing is identical; consequently, the effect of chunking the text would be not significant.

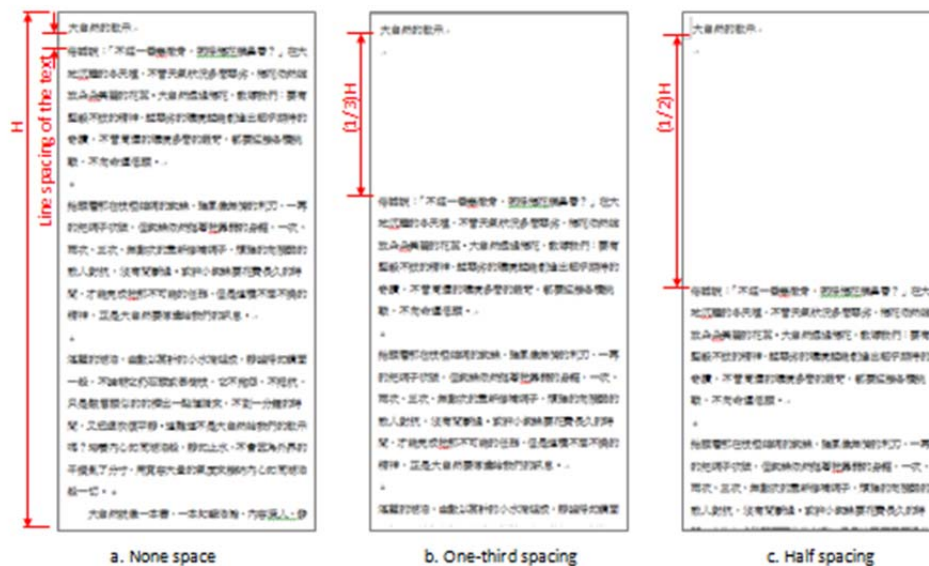


Figure 2. Three Levels of Page Spacing in the experiment.

The page spacing included three levels: none spacing, one-third spacing and half spacing shown on Figure 2. None spacing refers to the distance between the title and the main text is equal to the height of text line spacing; one-third spacing refers to the distance between the title and the main text is equal to the one-third height of the screen; half spacing refers to the distance between the title and the main text is equal to the one-half height of the screen.

Twelve essays were selected in the test. Subjects performed reading tasks to find the typos from the essays, and the reading time for each essay was recorded as a performance measure in the experiment. Both subjects and essays were two blocks in the Latin square design. This study also discussed subjective preferences for each typographic design.

Subjects

Twenty four undergraduate students from the Medical Colleges in Taiwan were recruited to participate in the experiment. Each subject performed only one experimental run. These subjects had attended the Chinese test of the General Scholastic Ability Test (GSAT) in Taiwan. The scaled score ranges from 0 to 15. These students had scores over 14 in the test. It means that these students had a good Chinese reading competence. They were paid \$12 USD for their participation in the experiment. Their mean age was 21.21 yr. (SD = 1.74). All subjects reported normal ocular health and no clinically significant anomalies of accommodation. They had 0.8 corrected visual acuities or better, and normal color vision.

All subjects consented to perform the trials by signing the agreements outlined by the committee on Governance Framework for Human Research ethics of National Cheng Kung University in Taiwan.

Materials

A 5.5-inch smartphone, modeled as ZenFone 2 Laser (ZE550KL), was used to display the reading comprehension test. Its display resolution is 267 ppi; and the screen size is 6.85cm X 12.18cm. Its physical size is 148.2cm x 72.8 x 10.34 mm. The experiment was programmed with Android 5.0 (Lollipop) on the smartphone to present the design combinations and control the experiment process and data collection.

Twelve essays were selected from the reading comprehension tests of a primary school in Taiwan. Ten typos were made in each essay on purpose. The original length of these essays was almost close to 501 characters. To get an identical length of essays, these articles were slightly modified. Finally, all the essays, including punctuation marks, had 501 characters in length.

The text font was a 12-point Hei style of Traditional Chinese characters. One point (pt) is equal to 1/72 inches and each Chinese character is confined to a box-shaped area. Therefore, the height of the box-shaped area for 12pt font is 4.2mm. However, the actual sizes measured from the screen directly are 2.8mm.

Tasks and Procedure

Subjects were asked to perform reading comprehension tests after they consented to perform the experiment. At the start of the trial, a “To start time counting” button (written with Chinese) was presented on the screen shown in Figure 3. Subjects were reminded to take a breath and ready to go before they tapped on the button. After they tapped on the button, the screen showed an essay with one of design combinations, and the start time was recorded by the programmed smartphone. Subjects were asked to find the typo in the essays as quickly as possible. After 10 typos were found, the experiment program recorded the ending time of the treatment combination. The reading time was assigned as ending time minus starting time.

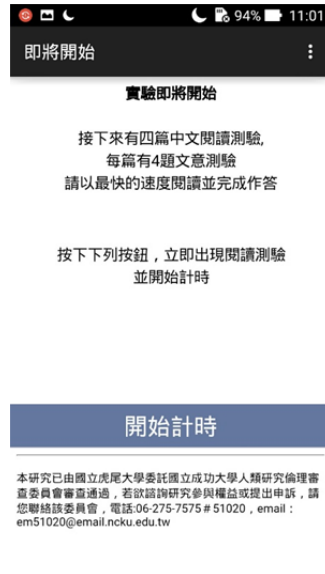


Figure 3. The preparing page has a blue bar, “to start time counting”. The words inside the button is written with Traditional Chinese; it means “to start time counting”.

At the same time, a subjective rating page with four questions showed out as Figure 4. The first question is to ask participants to rate their preference for typography of the read article. It included five choices, “prefer most”, “prefer”, “no opinion”, “less prefer” and “least prefer”, which were scored as “5”, “4”, “3”, “2”, and “1” respectively. The higher the score was, the more the participants preferred.

The second question is to rate aesthetic appealing of typography of the read article. It included five choices, “prefer most”, “prefer”, “no opinion”, “less prefer” and “least prefer”, which were scored as “5”, “4”, “3”, “2”, and “1” respectively. The higher the score was, the more beautiful the treatment combination was.

The third question asked subjects to evaluate their effort on reading the article. It included five choices, “very not heavy”, “not heavy”, “no opinion”, “heavy t” and “very heavy”, which were scored as “5”, “4”, “3”, “2”, and “1” respectively. The higher the score was, the less effort the subjects did.

The forth question asked subjects to evaluate their comfort on reading the article. It included five choices, “very comfort”, “comfort”, “no opinion”, “not comfort” and “very not comfort”, which were scored as “5”, “4”, “3”, “2”, and “1” respectively. The higher the score was, the more comfort the subjects felt.

美感評量

你喜歡這篇文章版面編排方式嗎?

☐ 非常喜歡

☐ 很喜歡

☒ 尚可

☐ 不喜歡

☐ 非常不喜歡

這篇文章版面編排方式好看嗎?

☐ 非常好看

☐ 很好看

☒ 尚可

☐ 不好看

☐ 非常不好看

回文章看版面 休息 下一篇文章

本研究已由國立虎尾大學委託國立成功大學人類研究倫理審查委員會審查通過，若欲諮詢研究參與權益或提出申訴，請您聯絡該委員會，電話：06-275-7575 # 51020，email：em51020@email.ncku.edu.tw

非常不好看

這篇文章的編排 讓你讀起來很吃力嗎?

☐ 一點也不吃力

☐ 有點 不吃力

☐ 尚可

☐ 有點 吃力

☐ 很 吃力

這篇文章的編排 讓你的眼睛感到很不舒服嗎?

☐ 一點也不會不舒服

☐ 有點 不會不舒服

☐ 尚可

☐ 有點 不舒服

☐ 很 不舒服

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Figure 4. Subjective rating page with four questions.

After finishing the questionnaire, subjects tapped on “go next article” button, and the next article appeared. Repeating the process of previous finding-typo test until subjects completed to read the twelve articles. Subject had 5 minutes break after finishing every two finding-typo tests. It spent about two hours for each subjects.

Familiarization was based on three training trial with an additional essay with different typographic variables.

Performance measures and data analysis

Reading time, a performance measure, was defined as the ending time minus the starting time in performing finding-typo tests. Analysis of covariance was employed to analyze the data. All calculations were made using the SPSS.

Besides, for subjective preference, the higher the score was, the more the participants preferred. As for aesthetic appealing of typography, the higher the score was, the more beautiful the treatment combination was. Next, for evaluation of effort on reading the article, the higher the score was, the less effort the subjects did. Finally, for evaluation of comfort on reading the article, the higher the score was, the more comfort the subjects felt.

3. Results

Table 1 shows the ANOVA Table of Reading Time for Testing Treatments. Results of analysis of variance indicated that the main effect of Page Spacing was significant ($F(2, 242)=3.49, p=0.032$); the main effect of Line Spacing was also significant ($F(1, 242)=5.68, p=0.018$); and the main effect of Paragraph Spacing was significant ($F(1, 242)=9.95, p=0.002$), too. However, the interactions of those factors were not significant.

Table 1. Results of ANOVA performed on the average reading times

Source	SS	df	MS	F	p-value
Treatment	199067.55	11	18097.05	2.25	0.0256
White Spacing (WS)	56179.51	2	28089.75	3.49	0.032
Line Spacing (LS)	45691.64	1	45691.64	5.68	0.018
Paragraph Spacing (PS)	80124.72	1	80124.72	9.95	0.002
WS X LS	9410.54	2	4705.27	0.58	0.558
WS X PS	373.83	2	186.91	0.02	0.977
LS X PS	5546.68	1	5546.68	0.69	0.407
WS X LS X PS	1740.65	2	870.33	0.11	0.898
Subject	1221460.71	22	55520.94	6.90	0.000
Essay	419086.69	11	38098.79	4.73	0.000
Replicate	38723.59	1	38723.59	4.81	0.038
Error	1947787.55	242	8048.71		
Total Error	4025193.65	287			

Table 2. Means and Standard Deviations of Reading Time for Each Level of Independent Variables.

Variables	Levels	n	Mean	S.D.	Duncan(0.05)	
Page Spacing	0	96	212.78	117.07	A	B
	1/3H	96	193.11	107.46	A	
	1/2H	96	227.19	128.56		B
Line Spacing	1 Char	144	198.43	122.59	A	
	2 Char	144	223.62	113.14		B
Paragraph Spacing	1 L-spacing	144	227.71	130.341	A	
	2 L-spacing	144	194.35	102.952		B

Note: Different letters in Duncan group indicate significant differences at 0.05 level.

Post hoc analysis was conducted. Table 2 shows the means and standard deviations of reading time for each level of the independent variables. Multiple comparison methods using contrasts were also used to test the difference of means of levels. The result showed no significant difference of reading time between 0H Page Spacing (212.78 sec) and 1/2 H Page Spacing (227.19 sec); and between 0H Page Spacing (212.78 sec) and 1/3H Page Spacing (193.11 sec). However, the reading time of 1/2H Page Spacing (227.19 sec) was significantly longer than 1/3H Page Spacing (193.11 sec). It suggested that oversized Page Spacing would degrade reading performance. Additionally, the result also showed the reading time for Line Spacing with 2-Charater (223.62 sec) was significantly longer than that with 1-Charater (198.43 sec). Besides, the result also showed the reading time for Page Spacing with 1-Line spacing (227.71sec) was significantly longer than that

with 2-Line spacing (194.35 sec).

Table 3 shows the ANOVA Table for subjective effort ratings. Results of analysis of variance indicated that the main effect of Page Spacing was significant ($F(2, 242)=9.861, p<0.01$). However, the main effect of Line Spacing was not significant ($F(1, 242)=1.774, p=0.184$); and the main effect of Paragraph Spacing was not significant ($F(1, 242)=0.350, p=0.555$), either. Besides, the interactions of those factors were not significant.

Table 3. Results of ANOVA for subjective effort ratings

Source	SS	df	MS	F	p-value
Page Spacing (WS)	12.507	2	6.253	9.861	.000
Line Spacing (LS)	1.125	1	1.125	1.774	.184
Paragraph Spacing (PS)	.222	1	.222	.350	0.555
WS X LS	1.715	2	.858	1.352	.261
WS X PS	.125	1	.125	.197	.657
LS X PS	.271	2	.135	.214	.808
WS X LS X PS	.396	2	.198	.312	.732
Subject	5.111	11	.465	.733	.707
Essay	108.177	22	4.917	7.754	.000
Replicate	2.000	1	2.000	3.154	.077
Error	153	242	0.634		
Total Error	285.111	287			

Table 4 shows the means and standard deviations of subjective effort scores for each level of Page Spacing. The high effort score means low effort. Multiple comparison methods using contrasts were also used to test the difference of means of levels. The result showed subjective score of 0H Page Spacing (4.06) was the highest; 1/3H Page Spacing(3.80) next; and 1/2 H Page Spacing (3.55) least. It suggested that oversized Page Spacing would increase reading effort.

Table 4. Means and Standard Deviations of Subjective Effort ratings for Each Level of Page Spacing

Dep. Variable	Effort	N	Mean	S.D.	Duncan(0.05)	Duncan(0.01)
White Space	0	96	4.06	0.86	A	A
	1/3H	96	3.80	0.99	B	A B
	1/2H	96	3.55	1.07	C	B

Table 5 shows the ANOVA Table for subjective comfort ratings. Results of analysis of variance

indicated that the main effect of Page Spacing was significant ($F(2, 242)=11.514, p<0.001$); and the main effect of Line Spacing was also significant ($F(1, 242)=9.533, p=0.002$). However, the main effect of Paragraph Spacing was not significant ($F(1, 242)=0.026, p=0.871$). Besides, the interactions of those factors were not significant.

Table 5. ANOVA Table for Subjective Comfort Ratings

Source	SS	df	MS	F	p-value
White Spacing (WS)	12.111	2	6.056	11.514	.000
Line Spacing (LS)	5.014	1	5.014	9.533	.002
Paragraph Spacing (PS)	.014	1	.014	0.026	.871
WS X LS	2.694	2	1.347	2.562	.079
WS X PS	.125	1	.125	0.238	.626
LS X PS	.528	2	.264	0.502	.606
WS X LS X PS	.083	2	.042	0.079	.924
Subject	7.486	11	.681	1.294	.228
Essay	135.972	22	6.181	11.751	.000
Replicate	.347	1	0.347	0.660	.417
Error	127.278	242.000	0.526		
Total Error	291.653	287			

Table 6. Means and Standard Deviations of Subjective comfort for Each Level of Page Spacing

Dep. Variable	Comfort	N	Mean	S.D.	Duncan(0.05)	Duncan(0.01)
Page Spacing	0	96	4.15	0.81	A	A
	1/3H	96	3.94	1.00	A	AB
	1/2H	96	3.65	1.13	B	B
Line Space	1 Char	144	3.78	1.05	A	
	2 Char	144	4.04	0.95	B	

Table 6 shows the means and standard deviations of subjective comfort scores for each level of Page Spacing. The high effort score means high comfort. Multiple comparison methods using contrasts were also used to test the difference of means of levels. The result showed both subjective scores of 0H Page Spacing (4.15) and 1/3H Page Spacing (3.94) did not significantly differ, but both subjective scores were higher than 1/2 H Page Spacing (3.65). Besides, the result indicated that the comfort scores for Line Spacing with 2-Character (4.04) was significantly higher than that with 1-Character (3.78).

Table 7 shows the ANOVA Table for subjective comfort ratings. Results of analysis of variance

indicated that the main effect of Page Spacing was significant ($F(2, 242)=20.404, p<0.001$); and the main effect of Line Spacing was also significant ($F(1, 242)=4.694, p=0.031$). However, the main effect of Paragraph Spacing was not significant ($F(1, 242)=0.522, p=0.471$). Besides, the interactions of those factors were not significant.

Table 7. ANOVA Table for Subjective Appealing Scores

Source	SS	df	MS	F	p-value
White Spacing (WS)	22.007	2	11.003	20.404	.000
Line Spacing (LS)	2.531	1	2.531	4.694	.031
Paragraph Spacing (PS)	.281	1	.281	0.522	.471
WS X LS	.646	2	.323	0.599	.550
WS X PS	.087	1	.087	0.161	.689
LS X PS	.563	2	.281	0.522	.594
WS X LS X PS	.840	2	.420	0.779	.460
Subject	3.205	11	.291	0.540	.875
Essay	101.465	22	4.612	8.552	.000
Replicate	.031	1	.031	0.058	.810
Error	130.507	242	0.539		
Total Error	262.163	287			

Table 8. Means and Standard Deviations of Subjective Appealing scores for Each Level of Independent Variables

Dep. Variable	beauty	N	Mean	S.D.	Duncan(0.05)	Duncan(0.01)
Page Spacing	0	96	3.92	0.74	A	A
	1/3H	96	3.58	0.87	B	B
	1/2H	96	3.24	1.11	C	C
Line Space	1 Char	144	3.49	0.99	A	
	2 Char	144	3.67	0.91	B	

Table 8 shows the means and standard deviations of subjective appealing scores for each level of Page Spacing. The high appealing score means high aesthetics. Multiple comparison methods using contrasts were also used to test the difference of means of levels. The result showed subjective score of 0H Page Spacing (3.92) was the highest; 1/3H Page Spacing (3.58) next; and 1/2H Page Spacing (3.24) least. It suggested that oversized Page Spacing would decrease subjective comfort. Besides, the result indicated that the appealing scores for Line Spacing with 2-Character (3.67) was significantly higher than that with 1-Character (3.49).

Table 9 shows the ANOVA Table for subjective preference ratings. Results of analysis of variance indicated that the main effect of Page Spacing was significant ($F(2, 242)=25.565, p<0.001$). However, the main effect of Line Spacing was not significant ($F(1, 242)=2.534, p=0.113$); Paragraph Spacing was also not significant ($F(1, 242)=0.144, p=0.705$). Besides, the interactions of those factors were not significant.

Table 9. ANOVA Table for Subjective Preference Ratings

Source	SS	df	MS	F	p-value
White Spacing (WS)	30.896	2	15.448	25.565	.000
Line Spacing (LS)	1.531	1	1.531	2.534	.113
Paragraph Spacing (PS)	.087	1	.087	0.144	.705
WS X LS	.715	2	.358	0.592	.554
WS X PS	.087	1	.087	0.144	.705
LS X PS	.563	2	.281	0.465	.628
WS X LS X PS	.465	2	.233	0.385	.681
Subject	5.677	11	.516	0.854	.586
Essay	101.715	22	4.623	7.651	.000
Replicate	.003	1	.003	0.006	.940
Error	146.229	242	0.604		
Total Error	287.969	287			

Table 10. Means and Standard Deviations of Subjective Preference for Each Level of Page Spacing

Dep. Variable	preference	N	Mean	S.D.	Duncan(0.05)	Duncan(0.01)
Page Spacing	0	96	3.90	0.75	A	A
	1/3H	96	3.48	0.87	B	B
	1/2H	96	3.09	1.18	C	C

Table 10 shows the means and standard deviations of subjective appealing scores for each level of Page Spacing. Multiple comparison methods using contrasts were also used to test the difference of means of levels. The result showed subjective score of 0H Page Spacing (3.90) was the highest; 1/3H Page Spacing(3.48) next; and 1/2 H Page Spacing (3.09) least. It suggested that oversized Page Spacing would decrease subjective preference.

4. Discussion

Like present study, previous studies asked participants to read articles with various typographic variables to verify the effects of those variables on readability. Both objective measures, Reading

time and Reading comprehension, were used to test the readability of text-reading frequently. Huang (2017) concluded that reading time was a suitable performance measure to measure reading performances, but not comprehension tests. Previous studies found no significant effect of typographic variables on reading comprehension when they performed multiple-choice comprehension tests (Wang and Chen, 2003; Chen and Chien, 2005; Chan and Lee, 2005). The reasons why the effects of typographic variables were not significant might be that the reading articles selected from the reading tests for high or primary school students were much easy to high educated participants who were college students with excellent linguistic competencies. Consequently, the effects of typographic variables were masked because all participants could get high comprehension scores easily when reading these materials. Hence, the scores of comprehension tests might be not sensitive enough to determine article readability without limiting reading time.

Inversely, the time spent on comprehension test have significant differences between different typographic variables (Chan and Lee, 2005; Cai et al, 2001; Moret-Tatay and Perea, 2011; Chan and Ng, 2012). It seemed that the reading time was sensitive enough to distinguish the differences between typographic variables. Therefore, in this paper, reading time was used as an objective performance measure. The outcomes indicated the effects of Page Spacing, Line Spacing and Paragraph spacing were significant on reading time.

Line Spacing

The result indicated that Line Spacing effect was significant on reading time. Wang et al. (2009) explored the effect Line Spacing including four levels, 1/4, 2/4, 3/4 and 4/4 character heights, which were not larger than 1 character height, on reading time. They found no significant effect of line spacing on reading time. Unlike the study of Wang et al. (2009), the present study investigated the effect Line Spacing whose sizes were not less than 1 character height, and found that the reading time for Line Spacing with 1-Character height was significantly less than that with 2-Character height. Both findings might be concluded that the reading speed would be degraded when the Line spacing size was larger 1-Character height.

Furthermore, previous studies of line spacing effects on desktop displays demonstrated that wider enough line spacing led to better reading performance. They found that double line spacing produced faster reading times and higher preferences than single line spacing (Ling and Schaik, 2007; Chan and Lee, 2005; Alan. et al., 2014). Obviously, the outcomes of present study, exploring white space effects on mobile phones, did not support those findings on large screens. The result demonstrated that line spacing with 1-character height had better reading performance than 2-character height. It seemed that participants preferred different line spacing sizes for different size of screens. For large screens (i.e. desktop displays), they preferred read the text with double character-height line spacing; however, for small screens (i.e. mobile phones), they preferred read the text with one character-height line spacing.

The inconsistent outcomes might result from their display sizes of reading texts. Due to the limit of display size of mobile phones, the text on the small display would overflow into multiple screens. Unlike using desktop displays, users have to slide their fingers to scroll the screen of smart phones in reading all the time. Previous studies believed that scrolling text might negatively affect performance (Morrison & Duncan, 1988; Piolat et al., 1997; Sanchez and Wiley, 2009; Sanchez and Goolsbee, 2010) because it would change the locations of all characters in the text, resulting in difficulty for users in locating information by recalling their spatial locations (Morrison & Duncan, 1988). In small screens, the length of an article with 2-character height line spacing would almost be twice the length of that with 1-character height line spacing. Consequently, participants had more sliding behaviors to scroll the text with 2-character height line spacing than that with 1-character height line spacing. Therefore, the result showed that participants had to spend more time to read an article with 2-character height line spacing than 1-character height line spacing.

Besides, the present paper also tested the effect of line spacing on aesthetic appealing. The result demonstrated that Line Spacing with 2-Charater height was more appealing than that with 1-Charater height. It was found that participants preferred Line Spacing with 2-Charater height to 1-Charater height in spite of they can read text with Line Spacing of 2-Charater height much faster than 1 Line Spacing of 2-Charater height. It conformed that aesthetic appearance might not always coincide with usability, even might degrade usability (Norman, 2002).

Based on his design experiences, Keyes (1993) proposed that the low tonality would increase text readability. However, the outcome did not show the advantages of low tonality on readability; but show subjects' appealing preference to low tonality. Hence, the proposition that the low tonality would increase text readability seemed to indicate the existence of a "what is beautiful is usable" stereotype that paralleled the work by Dion, Berscheid and Walster(1972) on the influence of aesthetics on perceptions in social settings.

In addition, the present paper also tested the effect of line spacing on comfort. Like aesthetic appealing, the result showed that Line Spacing with 2-Charater height was felt more comfortable than that with 1-Charater height. It also seemed to indicate the existence of a "what is beautiful is usable" stereotype. The judgment of comfort was influenced by judgment of beauty.

Paragraph Spacing

Paragraph Spacing would chunk the texts. The paragraph spacing size also included two levels: one line-spacing height, and double line-spacing height in the experiment. The size between paragraph spacing and line spacing would be identical when the paragraph spacing was 1 line-spacing-height; and the size of paragraph spacing would be twice of the line spacing when the paragraph spacing was 2-line-spacing height.

In another word, the text was chunked into discriminable blocks when Paragraph spacing was 2-line-spacing height; the paragraph blocks were not discriminable when Paragraph spacing was one line-spacing height.

The result indicated that the reading time of double line-spacing height was significantly slower than one line-spacing height. That is, subjects would spend less time to read the text which was chunked into discriminable

Increased Sliding behaviors resulted from increased text length would result in degraded readability.

1-line-spacing height increased less text length than 2-line-spacing height. However, the outcomes did not demonstrate that 1-line-spacing height had less reading time than 2-line-spacing height. The reason might be that the increased text length of 2-line-spacing height was not large enough to significantly increase more sliding behaviors than 1-line-spacing height. Therefore, Sliding behaviors did not significantly affect reading performance.

Inversely, 2-line-spacing height would chunk the paragraphs into discriminable blocks. Chunking would reduce the typographic tonality and facilitate reading comprehension (Keyes, 1993). Consequently, 2-line-spacing height of **Paragraph spacing** may enhance the overall readability and flow of the text. Therefore, the result demonstrated that the reading time of double line-spacing height was significantly slower than one line-spacing height.

Page Spacing

The result demonstrated that oversized Page Spacing would increase reading time; that is, degraded reading performance. On one hand, Page Spacing would increase Sliding behaviors resulted from its increased text length, which would result in degraded readability. Therefore, the outcomes showed that subjects have to increase their reading time when text was typeset with 1/2H Page Spacing. In another hand, Page Spacing would also provide relief to readers in reading a tedious text, and reduces loading of the reading task. Hence, the negative effect of sliding behaviors might be diluted with relief white space. Therefore, the outcomes did not show the difference of reading time between 0H Page Spacing and 1/3H Page Spacing.

However, subjective appealing and preference ratings also demonstrated that 0H Page Spacing was the most preferred; 1/3H Page Spacing next; and 1/2 H Page Spacing least. It suggested that oversized Page Spacing would decrease subjective preference; that is, white space would degrade subjective preference.

Design practitioners promoted the use of white space for elegance and ensuring a quality user experience. They believed that white space is a great tool to balance design elements and better organize content to improve the visual communication experience. However, many people without design background considered white space wasted space. They believed it should house more information or other visual elements to avoid wasting space. That is the reason why white space would degrade subjective preference.

The phenomena could be also generalized to subjective comfort and effort rating. The result showed both subjective sores of 0H Page Spacing and 1/3H Page Spacing did not significantly

differ, but both subjective scores were higher than 1/2 H Page Spacing. Subjects did not prefer 1/2H Page space because they considered white space wasted space. However, in the other, appropriate size of Page spacing would provide relief to readers to read the text. Hence, 1/3H page spacing could provide more comfort and less effort for subjects than 0H page spacing. Consequently, the difference of rating scores between 1/3H page spacing and 0H page spacing was not significantly different because the extend of unacceptable wasted space of white space might be diluted with relief of white space.

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佐證資料 1



國科會補助專題研究計畫項下出席國際學術會議心得

報告

日期：106 年 5 月 20 日

計畫編號	MOST 105-2221-E-150-034		
計畫名稱	小型手持裝置螢幕之版面留白對閱讀績效和美感喜好的影響		
出國人員姓名	黃室苗	服務機構及職稱	國立虎尾科技大學 教授
會議時間	106 年 5 月 13 日至 106 年 5 月 17 日	會議地點	日本 札幌
會議名稱	(中文)應用系統創新國際研討會 (英文) International Conference on Applied System Innovation (ICASI 2017)		
發表論文題目	(中文) 智慧型手機螢幕之正體中文字之文字尺寸和字體對閱讀績效的影響 (英文) Effects of Traditional Chinese Character Size and Font Style on Reading Performance When Using Smartphones		

一、參加會議經過

本人由 Google 的網站搜尋國際研討會時，發現 ICASI2017 是 IEEE 所授權的研討會，其討論之領域與本人的研究領域相同，並且，該研討會接受發表之論文將被收錄在 EI 索引的資料庫中，應是一個不錯的研討會，所以本人決定將研究成果發表於該研討會。後來，本人投稿發表的論文被該研討會大會所推薦，以 expending 文章發表

於 Applied Sciences 期刊(Impact Factor: 1.726 (2015) ; 5-Year Impact Factor: 1.773

(2015))，但轉投期刊需再支付投稿費用，所以本人經過思考後，自認該論文的研究品質和貢獻，應亦能為不用收取出版費的知名期刊所接受，所以為了為國家省下經費開支，本人決定不轉投該 SCI 期刊。目前本人正將此研究改寫為期刊格式，準備轉投 Applied ergonomics。

此次 ICASI2017 是由台灣知識創新設計學會和日本及中國等國家之大學合辦，自 5 月 13 日至 5 月 17 日共五天，包含 14 場 regular sessions 及 15 場 invited sessions，地點在札幌 Emisia Hotel，超過 600 篇論文發表，議題函蓋工業設計、創新設計、管理工程，材料工程、通訊工程、資訊技術、電子工程、綠色技術、文化研究、應用數學和其它。

第一天和最後一天分別是開幕式和閉幕式，精彩的活動是自第二天上午的大會主題演講(keynote speech)後開始，接著是密集的會議論文發表。本人參加的場次是”Industrial design and creative design”，於 5/16 日上午 11:00-12:00 發表，發表題目是本人科技部補助計畫，論文名稱為” Effects of Traditional Chinese Character Size and Font Style on Reading Performance When Using Smartphones”研討會論文將全部收錄在 IEEE 之論文集，可由 EI 資料庫查尋。國際會議的規模很大，每個時段只有 5 或 6 個發表會場，全部 6 個時段，全場討論熱烈，皆無冷場，各個都參與討論，提供正向意見和批判，使本人獲益良多。

二、與會心得

此次大會邀請三位學者為大會演講，其中，日本北海道大學的教授的演講內給本人有較大的啟發，他認為在這個大數據的時代，資料的收集似乎已不是關鍵的問題，但却伴隨著更多的人類解讀和消化訊息的能力。人們的能力有限，無法消化數以億計的資訊，所以他認為圖表的設計或是資訊視覺化的設計，在這個時代，它的重要性會持續增加。圖象認知一直是本人的研究領域，也是我所興趣的領域，他的見解，更顯得本人研究領域的重要。

在口語發表方面，此次研討會只有少數幾篇論文的作者是西方學者，其它都是東方人，以台灣學者最多。其中有不少之研究雖不成熟，但却給人有高度的啟發。例如，其中有一篇發表論文探討將工業設計的研究方法應用在牙醫作業空間的設計，經由 SCAMPER 的啟發找出設計解決方案，最終提出不錯的設計案。

另外，則是有一位博士生認為現在的設備，若增加感測元件時，可能需要重新設計機板，或重新修改程式。她則是解決此問題，只要把他的元件插入其母板，即可增加其感測數量。她聲稱，他所開發出來可擴增的模組有 11 個以上之多。

另外，有一位博士生發表以 CG network simulation，結合 GPS，利用智慧型手機記錄以影像記錄交通情況。他以程式設計方式，由 GPS 定位橋上或是橋下的影像，甚至能區別隧道內的影像。然而這個系統只能用於高速公路，對於道路混亂的台灣一般道路無法適用，很可惜。

另一位學者則是利用 TRIZ 的矛盾矩陣，以逆向工程的概念創新專利回避的方法，

他的程式可以檢索國內的專利，若用到大國的專利系統，貢獻更大。

最後，我發現到，雖然研討會已逐漸成為學生論文發表的練習場，資深研究學者無法實質由研討會中有更多的啟發，造成愈來愈少的學者實質參加，僅僅指派學生發表；但，在這個研討會中，我發現，這些被指派發表的博士生，表現傑出。也就是說，台灣年輕學者的程度一直在進步，顯現台灣研究能力的不可輕忽性。

三、建議

此研討會的內容，著實給予本人諸多的啟發，明年應再積極參加此研討會，也應鼓勵更多學者參與，互相交流進步。

四、攜回資料名稱及內容

1. 會議時程手冊 一本。
2. 發表證明書一張。

論文被接受發表之大會證明文件



May 13-17, 2017, Hotel exelle, Sapporo, Japan <http://2017.icasia.org/>

Receipt

Paper No.:0032

Paper Title: Effects of Traditional Chinese Character Size and Font Style on Reading Performance When Using Smartphones

Register author: Shih-Miao Huang

Affiliation: National Formosa University

Dear Prof. Shih-Miao Huang,

This is to acknowledge that we have received the registration fee, total amount paid: **NTD 15,500**, for the submission referenced above, the 2017 IEEE International Conference on Applied System Innovation (IEEE ICASI 2017).

Prof. Ardis Donald Kin-Tak Lam, Ph.D
IEEE ICASI Program Chairman



Prof. Teen-Hang Meen, Ph.D
TIKI President



Taiwanese Institute of Knowledge Innovation

發表證明



發表論文

Format Effects of Traditional Chinese Character Size and Font Style on Reading Performance When Using Smartphones

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Abstract

Previous studies which explored the effects of typographic variables on Chinese text reading on small devices fixed the viewing distance from eyes to the screens. Hence, their results might not be generalized to the typographic design of Chinese texts on small devices. Therefore, the article tried to investigate the effects of Traditional Chinese character size and font style on reading performance when subjects used 5.5-inch smartphones without limiting their viewing distances. The experiment a 3(character size) $\times$ 3(typeface) completely randomized replicate design. Each design combination was tested twice. Each subject read only one of the experimental runs to perform reading comprehension with following multiple-choice tests. The reading time was a performance measure. Analysis of covariance was employed to analyze the data. Unlike previous studies, the results found that the font size did not significantly affect reading performance because subjects could adjust their viewing distance freely in real situations, resulting the legibility of various character sizes did not significantly differ. In addition, the font style effect was significant. The results showed that readers had better reading performance when the texts were written with the Kai style when using smartphones. It seemed to indicate that the serifs fonts might be a good element to recognize a Chinese character easily. Finally, the results also implied that the scrolling effect did not affect reading time significantly.

Keywords: Chinese text reading, typography, smartphone.

Introduction

Due to the convenience of handy carrying, users can use smartphones at any time, place or occasions. The population of reading either digital news or e-books on smartphones significantly continues to increase due to increasing penetration of smartphones. Some studies investigated how these typographic factors affect reading efficiency of Traditional Chinese text on the small screen because the need to read Traditional-Chinese text on small displays is increasing, (e.g. [5]). The effects of two Chinese typographic variables, font size, and typeface were frequently explored to discuss their effects on Chinese text reading performances.

A. Font Size

Previous studies found that Chinese characters with larger font sizes had better legibility [1] [2]. However, Chinese characters are visually more complex and typically have more

structural details within each character than the shapes of English letters. A relative larger visual angle should be allowed for Chinese characters. Cai et al. [3] found that all Chinese characters could be clearly recognized when their sizes were over 20.39' in 15-inch CRT color monitor. Wu [4] suggested the Chinese character sizes should be over 55.2' for aged people; 50.0' for the middle-aged; and 42.0' for young people in a 6.5-in electronic paper display.

However, character legibility is only a measure of how easily a reader can distinguish individual letters or characters from each other. Readability is more than simply legibility. It also has to consider the other factors, such as reading comprehension, reading speed, preference, fatigue, etc. The Chinese texts with 14 pt (38') are superior to that at 10 pt (27') in reading speed, reading comprehension and subjective preference when subjects read the text in 15-inch color CRT [5]. They also found that subjects reported less fatigue, more reading ease, more comfort when reading texts with larger font size.



Fig. 1 The four font styles of Chinese characters

Besides, for reading in mobile phones, Huang, Rau and Liu [6] also found that the reading speed of text in large fonts (ex. 26' visual arc) was also significantly faster than in small fonts. However, not all the scientific findings agreed that the text written with a larger font size had better readability than small characters. The text written with a larger font size might occupy more space and lengthens the text, thus increasing the number of saccades. Consequently, readers have to spend more time to read through the lengthened text in large font. That is, the text in smaller letters would conversely promote more reading performance than in larger letters when both the smaller characters and the larger characters are legible. all fonts (ex. 14' visual arc). Chan and Ng [7] found that the 14-point characters were rated significantly superior to the 12-point characters in a 17-inch LCD monitor.

However, not all the scientific findings agreed that the text written with a larger font size had better readability than small characters. The text written with a larger font size might occupy more space and lengthens the text, thus increasing the number of saccades. Consequently, readers have to spend more time to read through the lengthened text in large font.

That is, the text in smaller letters would conversely promote more reading performance than in larger letters when both the smaller characters and the larger the characters are legible.

It seems that the ideal font size optimizing readability should be the smallest size that could be recognized legibly and that eliminate the number of saccades to facilitate reading speed. To verify this assertion in Chinese text reading, Yen et al. [8] assigned both Chinese characters in 32 pixels (60' arc) and 24 pixels (41' arc) to texts in 19-inch CRT monitor, respectively. Both sizes are larger than the legible size for presenting Chinese (i.e. 38' arc suggested by Chan and Lee [5]). Their findings supported the assertion and found that the number of fixations for large characters (60' arc) is significantly larger than small characters (40' arc). The results also indicated that the total reading time was shorter for 24-pixel characters compared with 32-pixel characters.

Furthermore, previously mentioned studies above were conducted in large displays which were above 14 inches bigger than the size of handheld mobile devices. Sanchez and Goolsbee [9] found that character sizes might interact with display size during text-reading. They believed that texts with small characters could improve text-recall tasks performed on small displays, but not large display. Therefore, findings from above studies conducting experiments in desktop monitors (sized from 15 to 19 inches) might not be generalized to typographic design on the small screens of smartphones.

Besides, reading behaviors on desktop displays are little resemblance to those using handheld devices. The smartphone user could hold the phones and adjust his view distance in a comfortable reading posture by bending his elbow. To see the characters clearly, users could adjust their viewing distances to attain an optimal visual angle subtended to the retina. In this situation, the effects of font sizes on reading performances might be not significant.

Therefore, this article argued that the character size effect on reading performance might be not significant for held-hold displays on reading performance.

A. Font Styles

The effect of font styles on readability from a computer screen was investigated. Three Chinese typefaces, Kai, Ming, and Hei, are extensively used in computer-display text [3]. A Kai typeface resembles handwritten calligraphy shown in Fig. 1-a; the stroke width from start to end is not consistent and the stroke extremities are with curve shapes. A Ming typeface (Fig. 1-b), which has simpler stroke patterns for printing, is similar to Western serif typefaces such as Norman. A Hei typeface (Fig. 1-c), which has a consistent stroke width with minimalistic design aesthetics, is similar to Western sans-serif typefaces such as Verdana. Besides, a Li style (Fig. 1-d), a clerical script, was also discussed for its legibility on displays. The horizontal strokes of Li styles are wavy lines with flared tails. Therefore, Li style is more visually complicated than the other three styles. Besides, among the other three typefaces, Kai typefaces (with variant stroke width) are the most visually complicate; the Ming style (with decorated serifs) is the next, and the Hei typeface (with no decorated strokes) is the least complicated.

Previous studies showed a significant effect of typeface on legibility for Chinese text displayed on electronic displays. Their outcomes showed that the least complicated typeface (i.e.

Hei) was the most legible; the most complicated typeface (i.e. Kai, or Li) was the least legible; and the middle complicate typeface (i.e. Ming) was between them [3] [10] [11] [12].

Besides, previous studies did not find a significant effect of typeface style on Chinese text reading comprehension when reading time was not limited [5] [13] [14]. The reason might be that the reading materials were selected from the reading comprehension tests for high school students and they are too easy to the subjects who were college students.

However, the time of executing comprehension test would have significant differences between different typefaces. Chan and Lee [5] found the reading time of texts with Ming styles was significantly shorter than Li styles when conducting reading comprehension tests. They believed that Ming characters were more legible than Li characters because Ming characters are found to have larger height and density than Li characters when both had the same points [3]. Therefore, they concluded that the actual height of different styles with identical points might account the differences of reading performance. Those studies confined subject viewing distances to a fixed distance.

In addition, Chan and Ng [7] indicated that the Chinese characters with serifs had better performance on comparison-proofreading tasks than those with sans serifs when the characters were small. Possibly, Chinese characters belong to hieroglyphic systems. Their distinctive features are not to be obscure when the words are presented in less legible situations. Distinctive features refer to the critical parts of an object offer the information to recognize the object. Zimbardo and Gerrig [15] believed that one could recognize objects with limited information, but not if that information disrupted the critical parts. Therefore, Chinese characters with serifs had better performance on the text reading because the serifs may provide an extra cue to the location of letter strokes and make letters perceptually more unique and identifiable [16].

Therefore, the present study tries to verify two arguments. Firstly, do the small font sizes have better reading performance than larger ones when users read Traditional Chinese text on smartphones? Secondly, do the typefaces affect performances of Traditional Chinese text reading on smartphones?

Methods

The experiment a 3(character size) $\times$ 3(typeface) completely randomized design. Each design combination was tested twice. The character size had three levels: 10pt, 12pt, and 14pt. The typeface had three levels, including Hei, Ming, and Kai styles. Furthermore, the subject reading capability would affect reading performance. Hence, reading capability would be a covariate in the experiment. To calculate reading capability, each subject had to read an essay written with a 12pt-Li style to conduct reading comprehension test before performing experimental trials; and the reading time was recorded and regarded as a covariate.

A. Subjects

Eighteen undergraduate students from the Medical College in Taiwan were recruited to participate in the experiment. Their mean age was 20.83 yr. (SD = 2.60). Each subject performed only one experimental run. These students had attended the Chinese test of the General Scholastic Ability Test

(GSAT) in Taiwan. The scaled score ranges from 0 to 15. These students had scores over 14. It means that these students had good Chinese reading skills. They were paid \$12 USD for their participation in the experiment. All subjects reported normal ocular health and no clinically significant anomalies of accommodation. They had 0.8 correct visual acuities or better, and normal color vision. All subjects consented to perform the trials by signing the agreements outlined by the committee on Governance Framework for Human Research ethics of National Cheng Kung University in Taiwan.

B. Materials

Four essays were selected from the reading comprehension tests of a primary school. The lengths of the four essays, including punctuation marks, were 784, 728, 424 and 801 characters, respectively. The first essay was typeset with a fixed typography, i.e. 12-pt-Li style; the other three were designated to each treatment combination to test its reading performance. Four questions are showed at the end of each essay. Subject performed multiple-choice tests for each question with four possible answers.

A 5.5-inch smartphone, modeled as ZenFone 2 Laser (ZE550KL), was used to display the reading comprehension test. Its display resolution is 267 ppi; and the screen size is 6.85cm X 12.18cm. Its physical size is 148.2cm x 72.8 x 10.34 mm. The experiment was programmed with Android 5.0 (Lollipop) on the smartphone to control the experiment process and data collection. Due to the limitation of screen size, the text would overflow onto multiple screens. Subjects can scroll forward or backward to read the essays.

C. Tasks and Procedures

Subjects were asked to read the four essays to perform reading comprehension tests after they consented to perform the experiment. At the start of the trial, a “Start to the time counting” button (written with Chinese) was presented on the screen. Subjects were reminded to take a breath and ready to go before they tapped on the button. After they tapped on the button, the screen showed the essay with a 12-pt-Li style, and the start time was recorded by the programmed smartphone. Subjects read the essay and performed multiple-choice tests for four questions at the end of the essay. Subjects were told that they had to choose the right answers as possible as they could. After they complete the multiple-choice tests, they tapped the “finished” button at the end the essay, and the end time was recorded. The subject reading capability was defined as the reading time which was the end time minus the start time.

After they tapped on the “finished” button, the screen showed the three essays in sequence with one of nine typographic designs and the start time was recorded by the programmed smartphone. Subjects read the essays in sequence and performed multiple-choice tests for four questions at the end of each essay. Subjects were told that they had to choose the right answers as possible as they could. After they complete the multiple-choice tests, they tapped the “finished” button at the end the passages, and the end time was recorded.

Each subject spent less than ten minutes taking the test. Familiarization was based on one training trial with an additional essay with different typographic variables.

B. Performance Measure and Data Analysis

Reading time, a performance measure was defined as the ending time minus the starting time, with at least 80% correct responses for the multiple-choice tests. Analysis of covariance was employed to analyze the data. All calculations were made using the SPSS.

The reading essays were selected from the reading comprehensive tests of a primary school in Taiwan. Therefore, these articles were easy for all the subjects whose Chinese test scores of GSAT are over 14. Therefore, all the subjects responded 91% correct answers for the multiple-choice tests at least.

Results and Discussions

Table 1 shows the means and standard deviations of the reading time for each level of the independent variables. The results of analysis of covariance did not indicate a significant effect of Font size ($F(2, 8)=0.619, p<0.563$) in this study; however, the main effect of Font style was significant ($F(2, 8)=5.529, p=0.031$). Multiple comparisons showed that reading time for the Kai style was significantly shorter than both the Hei style and the Ming style. However, the reading time for the Hei style and the Ming style did not differ significantly. The results indicated that readers had better reading performance when the texts were written with the Kai style when using smartphones.

Previous studies found that the reading comprehension scores were not a useful indicator to find out the optimal typefaces because their essays used in the test were too easy to be understood by all participants. Therefore, in this paper, reading time was used as a performance measure. The outcomes did find a significant effect of type styles on reading performance. Chan and Lee [5] also found reading time of texts with different type styles was significantly different. The outcomes might suggest that reading time was a suitable variable to measure reading performances.

A. Legibility

Besides, the outcomes seemed to show that the legibility of Chinese characters did not a critical factor influence reading time. Previous studies believed that the more complicated the typeface was, the less legible it was. However, the outcomes did not show this phenomenon. Chan and Ng [7] believed the texts with Ming styles (less legible) were faster than Hei styles (more legible) only when the texts were written in small characters (i.e. 10pt in their study). There might be several reasons to explain why the legibility of Chinese characters did not influence the reading time. Firstly, the recognition of Traditional Chinese characters in text readings involves a top-down recognition process which believes that the text can provide a context that gives a perceptual set for an ambiguous figure (i.e. a Chinese character). Hence, subjects could know the characters intuitively, even the characters with high frequencies degrading visual recognition. Second, Traditional Chinese characters are kinds of logograms which refer to the use of graphics or similar shapes to express what they represent. Hence, the process of recognition of graphic-like Chinese characters also involves a global or holistic process in which a Chinese character is viewed in whole, but not viewed

its details.

On the other side, the reason why Kai characters were more easily to be read than Hei characters or Ming characters might be the complexity of their stroke widths. The extremities of Kai character strokes are curved. Consequently, the global feature of a Kai character is more distinctive from that of the other characters because its stroke curved ends afforded hints to recognize it.

B. Font Size

Previous studies found that texts written with the large character size had better reading performances than those written with small characters because the larger font size had better legibility on desktop screens. However, the results of the present article did not indicate a significant effect of Font size. This article focusing on smartphones was different from previous studies which performed experiments on the desktop computer. Therefore, the outcomes of previous could not be generalized to smartphone design.

Although, the larger characters with larger visual angles might have shown a better reading performance than the small characters. However, subjects could adjust their viewing distances from their eyes to the screen of the smartphone to get their own optimal visual angle of characters in the experimental trials. Consequently, the visual angles for various character sizes were not significantly different. This would be the reason why the results showed no significant effects of font size on reading performance.

C. Scrolling Effect

Large characters would lengthen text and overflow into the next pages although large characters are more legible than small characters. Subjects needed to increase their scrolling behaviors to read the characters overflowing into the next page. Previous studies believed that scrolling negatively impact reading performance on full-size displays because of the added difficulty in locating information within scrolling text [17]. Therefore, they believed that large characters would degrade reading performance. However, the result did not show the scrolling effect in this study. The text occupied almost 2.5 pages for the texts with 14pt font size; about 2 pages for the texts with 12pt font size; and about 1.5 pages for the texts with 10pt font size. It might be that the differences between the lengthened texts among these three font sizes were not large enough to arouse scrolling effects.

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TABLE I
MEANS AND STANDARD DEVIATIONS OF READING TIME
FOR EACH LEVEL OF INDEPENDENT VARIABLES.

	Level	N	Mean	S. D.	Groups
Font Style	Kai	6	404.45	109.35	A
	Hei	6	473.17	134.17	B
	Ming	6	481.98	113.74	B
Font Size	10	6	476.22	141.96	A
	12	6	442.78	126.13	A
	14	6	440.61	117.85	A

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105年度專題研究計畫成果彙整表

計畫主持人：黃室苗				計畫編號：105-2221-E-150-034-			
計畫名稱：小型手持裝置螢幕之版面留白對閱讀績效和美感喜好的影響							
成果項目				量化	單位	質化 (說明：各成果項目請附佐證資料或細項說明，如期刊名稱、年份、卷期、起訖頁數、證號...等)	
國內	學術性論文	期刊論文		0	篇		
		研討會論文		0			
		專書		0	本		
		專書論文		0	章		
		技術報告		0	篇		
		其他		0	篇		
	智慧財產權及成果	專利權	發明專利	申請中	0	件	
				已獲得	0		
			新型/設計專利		0		
		商標權		0			
		營業秘密		0			
		積體電路電路布局權		0			
		著作權		0			
		品種權		0			
		其他		0			
		技術移轉	件數		0		件
	收入		0	千元			
	國外	學術性論文	期刊論文		1	篇	正撰稿，預計發表於Applied Ergonomics
			研討會論文		1		發表於ICASI 2017(International Conference on Applied System Innovation (ICASI 2017))，佐證請參閱出國報告表
專書			0	本			
專書論文			0	章	發表於IEEE國際研討會一篇，並蒐錄於研討會的專書。		
技術報告			0	篇			
其他			0	篇			
智慧財產權及成果		專利權	發明專利	申請中	0	件	
				已獲得	0		
			新型/設計專利		0		
		商標權		0			
		營業秘密		0			

		積體電路電路布局權	0		
		著作權	0		
		品種權	0		
		其他	0		
	技術移轉	件數	0	件	
		收入	0	千元	
參與計畫人力	本國籍	大專生	0	人次	
		碩士生	0		
		博士生	0		
		博士後研究員	0		
		專任助理	0		
	非本國籍	大專生	0		
		碩士生	2		參與並學習實驗過程和數據分析
		博士生	0		
		博士後研究員	0		
		專任助理	0		
其他成果 (無法以量化表達之成果如辦理學術活動、獲得獎項、重要國際合作、研究成果國際影響力及其他協助產業技術發展之具體效益事項等，請以文字敘述填列。)			參加研討會，論文獲ICASI 2017 FIRST PRIZE Paper Award。請參 成果報告中 佐證資料1。		

科技部補助專題研究計畫成果自評表

請就研究內容與原計畫相符程度、達成預期目標情況、研究成果之學術或應用價值（簡要敘述成果所代表之意義、價值、影響或進一步發展之可能性）、是否適合在學術期刊發表或申請專利、主要發現（簡要敘述成果是否具有政策應用參考價值及具影響公共利益之重大發現）或其他有關價值等，作一綜合評估。

1. 請就研究內容與原計畫相符程度、達成預期目標情況作一綜合評估

☒ 達成目標

☐ 未達成目標（請說明，以100字為限）

☐ 實驗失敗

☐ 因故實驗中斷

☐ 其他原因

說明：

2. 研究成果在學術期刊發表或申請專利等情形（請於其他欄註明專利及技轉之證號、合約、申請及洽談等詳細資訊）

論文：☐ 已發表 ☐ 未發表之文稿 ☒ 撰寫中 ☐ 無

專利：☐ 已獲得 ☐ 申請中 ☒ 無

技轉：☐ 已技轉 ☐ 洽談中 ☒ 無

其他：（以200字為限）

3. 請依學術成就、技術創新、社會影響等方面，評估研究成果之學術或應用價值（簡要敘述成果所代表之意義、價值、影響或進一步發展之可能性，以500字為限）

目前的手機排版研究皆桌上型螢幕上執行，且依人因規範，設定450mm以上視距。然而手機的使用和桌上型電腦的使用相很大，甚至正常手持使用下，視距經常不足300mm。因此之前有關的排版研究中，所建議的字體大小、字型、行距和留白可能不合於手機之文字排版。在者，設計師大大都認為留白是重要的設計元素，然却少有針對留白所做的科學學研究。本研究為此，針對手機的操做，將實驗程式寫入手機，進行設計測驗。實驗結果果然不同於以往之建議。此一突破的研究成果，可供手機文字排版設計之應用。

4. 主要發現

本研究具有政策應用參考價值：☒ 否 ☐ 是，建議提供機關（勾選「是」者，請列舉建議可提供施政參考之業務主管機關）

本研究具影響公共利益之重大發現：☐ 否 ☐ 是

說明：（以150字為限）

1. 手機滑動效應是主要顯著影響文章閱讀的主因，所以少留白可減少文章太長，溢至下一版面，減少手滑手機，促進閱讀效率。但另一面，留白提高 tonality，造成瞻讀性不佳。

2. 留白型式會影響閱讀的時間和喜好度